

Eavesdropping on a conversation within these networks was easy as pie. All you needed was a scanner; once tuned into a frequency in the cellular range, you could sit back and listen away. First, they tried to manufacture scanners that wouldn't tune into these frequencies at all. Naturally, this didn't work. It was difficult to procure such a scanner (and really, who would go hunting for one?), and even if you did end up with this piece of machinery, there was nothing a little re-programming couldn't fix. The next idea was to 'scramble' the signals, de-scrambling them at the receiving end. Scrambled signals were still quite easy to tap into, but at least now the casual listener wouldn't be able to listen to private conversations. This is about where security stopped in these systems.

Another disadvantage of the analogue systems was the difficulty in transmitting data over them. Partially digitising the system made this a little less difficult, but it was still less efficient than the newer fully digital systems which were to follow.

Standing up—The Second Generation (2G)

The Second Generation of cellular technology was marked by a shift from analogue to digital systems.

Shifting to digital networks had many advantages. Firstly, transmission in the digital format aided clarity, since the digital signal was less likely to be affected by electrical noise. Secondly, transmitting data over digital network is much easier; data could also be compressed, saving a lot of time. And finally, with the development of new multiplexing techniques, the capacity of the cellular network could be increased manifold.

The technologies in a 2G cellular network are based on one of two concepts:

m Time Division Multiple Access (TDMA)

Just like FDMA separated calls by assigning them different frequencies, TDMA separated calls by assigning them different time slots in the same frequency. TDMA was used in conjunction with